Public Release Request for RadarData.jl and SARkit.jl

Matrix Research

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1 Description

This is a public release request for RadarData.jl and SARkit.jl.

RadarData.jl is a Julia package that provides data structures and utilities for handling synthetic aperture radar (SAR) data, designed to specify minimal data structures commonly used for radar processing at Matrix Research, Inc. The package focuses on core data types and does not contain any processing algorithms beyond standard range compression, nor does it contain system-specific implementations.

The core functionality centers around the Video Phase History (VPH) data structure, which represents range-compressed radar pulse data with a minimal set of associated platform positions, timing, and frequency information. The package also includes coordinate system transformations and utilities for querying metadata in both frequency and time domains.

SARkit.jl provides conversion functions between National Geospatial-Intelligence Agency (NGA) standard radar data formats and the Matrix Research data types defined in RadarData.jl. SARkit bridges the gap between government-standard formats (CPHD, SICD, SIDD) and Matrix Research's internal data structures, enabling interoperability with NGA-compliant radar processing tools and datasets. The package leverages Valkyrie Systems's open-source `sarkit` library through Julia's PythonCall interface to read and write these standardized formats.

By publicly releasing both RadarData and SARkit, we can establish an openly-documented interface compatible with proprietary Matrix Research software and hardware while providing integration with NGA standard formats. This creates a relatively lightweight radar data handling specification for the Julia programming language that benefits communication with both government and academic radar research communities, while safeguarding sensitive processing algorithms.

2 Function and Use of Software

2.1 RadarData.jl: Matrix Research Data Types

2.1.1 Video Phase History (VPH) Data Structure

The VPH (Video Phase History) data structure is the central component of RadarData. It stores complex-valued radar returns in a matrix format (frequency or fast-time $\times$ slow-time) along with associated metadata including:

- **freq_list_hz**: Frequency list in Hz for each range bin
- **ref_range_m**: Two-way range to center range bin per pulse
- **slow_time_s**: Per-pulse collection time
- **tx_pos_enu, rx_pos_enu**: Transmitter/receiver positions in East-North-Up coordinates
- **srp_lla**: Scene reference point in local and geodetic coordinates
- **range_domain**: Boolean flag indicating frequency vs. range (fast-time) domain

The VPH structure supports both monostatic and bistatic radar geometries and can work with data in either the frequency domain or range (fast-time) domain. Functions `vph_to_rs!()` and `rs_to_vph!()` provide efficient in-place transformations between these domains.

2.1.2 Position, Navigation, and Timing (PNT)

The PNT (Position, Navigation, Timing) module provides structures for storing and interpolating platform state information. The PVA (Position, Velocity, Attitude) structure seeks to comply with ASPN conventions and includes functions for interpolating states between measurements, computing ranges to scene points, and transforming between coordinate systems.

2.1.3 Package Extensions

The package uses Julia's extension system to provide optional support for converting other data types to RadarData types. These extensions allow RadarData to interface with various data sources without requiring all users to install specific dependencies. As an example, `SoulExt` provides access to data collected by Matrix Research's radio frequency system on a chip (RFSoC)-based radars.

2.2 SARkit.jl: NGA Format Conversions

SARkit.jl serves as a bridge between NGA standard radar data formats and the Matrix Research data types defined in RadarData.jl. The package provides conversion functions for the following NGA standard formats:

- **CPHD (Compensated Phase History Data)**: A standard format for storing phase history data with associated metadata and channel parameters
- **SICD (Sensor Independent Complex Data)**: A standard format for complex-valued SAR imagery with comprehensive metadata
- **SIDD (Sensor Independent Derived Data)**: A standard format for derived SAR products and display-ready imagery

SARkit uses Julia's PythonCall package to interface with the Python `sarkit` library, which handles the low-level parsing and validation of these XML-based formats. This approach allows SARkit to leverage mature Python tools for NGA format compliance while providing native Julia interfaces for working with RadarData structures.

The primary use case is reading NGA-format radar data and converting it to VPH and SARImage structures for processing with Matrix Research tools, as well as exporting processed results back to NGA-compliant formats for distribution and archival.

3 Potential End Users

We expect RadarData and SARkit to be primarily used by Matrix Research developers and their collaborators. We envision that providing a set of external-facing data standards will provide a core for future hardware and software efforts to target.

SARkit specifically enables interoperability with government and academic partners who work with NGA standard formats. By providing conversion tools between Matrix Research's data structures (RadarData) and widely-adopted government standards (NGA formats), we facilitate communication and make it simpler to use Matrix Research hardware and software products with existing radar processing workflows. This provides potential for wider adoption and minimizes friction when support for new processing or a new data source is requested.

Additionally, these packages provide support for radar signal processing developers in the Julia language. This language strikes a balance between ease of development and efficient execution that is desirable for the types of niche, often experimental, software developed for advanced radar signal processing.

4 Technical Specifications

4.1 Programming Language

Julia (version 1.12.4+)

4.2 Operating System

Cross Platform (Windows, Mac, Linux)

4.3 Other Relevant Features

Both RadarData and SARkit follow modern Julia development practices including:

- Continuous integration with testing
- Automatic documentation generation
- Type-stable implementations for performance
- Package extension system for optional dependencies
- In-place operations (!) for memory efficiency

SARkit additionally requires:

- Python runtime with the `sarkit` package installed

- PythonCall.jl for Julia-Python interoperability
- Dependencies for XML parsing (`lxml`) and numerical operations (`numpy`)

5 DoD Program

[To be filled in - program information not provided in source files]

6 System or Product Relationship

RadarData provides the foundational data structures used across multiple radar processing systems developed by Matrix Research. It serves as a common interface layer between radar data acquisition systems and signal processing algorithms.

SARkit extends RadarData's utility by providing conversion functions to and from NGA standard formats, enabling Matrix Research systems to integrate with government and commercial radar processing workflows that use CPHD, SICD, and SIDD formats. Together, these packages form a complete data handling solution that supports both internal Matrix Research data types and external standardized formats.